



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
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SDC 1-Day 7

**II-II CSE,
CSE(AI&ML) AY:
2026-2027**

SL No	Week	Part A	Detailed Subtopics	Part B	Detailed Subtopics
1	Week 3	Random Forest	Random Forest, Ensemble Learning, Bootstrapping, Bagging, Randomness in Random Forest, Working of Random Forest, Overfitting Reduction, Important Hyperparameters, Advantages, Limitations	Practical Implementation	Day7_Exercise7_Random_Forest_Wine

Contents:

- 1.What is Random Forest?
- 2.Why Do We Need Random Forest?
3. What is Ensemble Learning?
4. What is Bootstrapping?
5. What is Bagging?
6. Where Does the “Random” in Random Forest Come From?
7. How Does Random Forest Work?
8. Why Does Random Forest Reduce Overfitting?
9. Important Hyperparameters of Random Forest
- 10.Advantages of Random Forest
- 11.Limitations of Random Forest
- 12.Q & A

1. What is Random Forest?

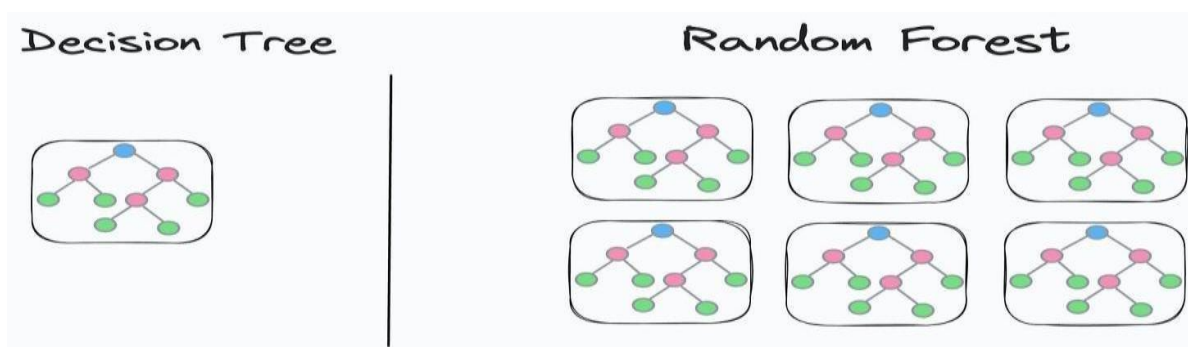
Random Forest is a **Supervised Machine Learning Algorithm** that uses multiple Decision Trees to make predictions.

Instead of relying on a single Decision Tree, Random Forest combines the predictions of many trees to produce a **more robust and reliable final prediction**.

Random Forest = Multiple Decision Trees + Combining Their Predictions

It can be used for:

- **Classification**
- **Regression**



2. Why Do We Need Random Forest?

A single Decision Tree can easily become **overfitted**.

For example:

- The tree may grow very deep.
- It may learn the training data too closely.
- A small change in the training data can create a very different tree.

This makes a Decision Tree a **high-variance model**.

Random Forest solves this problem by creating **multiple Decision Trees** and combining their predictions.

This generally helps to:

- Reduce overfitting
- Reduce variance
- Improve generalization
- Produce more stable predictions

3. What is Ensemble Learning?

Ensemble Learning is a machine learning technique where multiple models are combined to produce a better final prediction.

The basic idea is:

Multiple models working together can often produce a more robust prediction than a single model.

There are different ensemble techniques, such as:

- Bagging
- Boosting
- Stacking

Random Forest is mainly based on:

Bagging

4. What is Bootstrapping?

Bootstrapping is a sampling technique where multiple datasets are created from the original dataset by sampling with replacement.

Key Points

- Same data point can appear multiple times
- Some points may not appear at all
- Each sample is the same size as the original dataset
- Used to create diverse training sets

Bagging (Bootstrap Aggregating)

Bagging is an ensemble technique that:

- Trains multiple models on bootstrapped datasets
- Combines their predictions to produce a final output

5. What is Bagging?

Bagging stands for **Bootstrap Aggregating**.

It is the main ensemble technique used in Random Forest.

The process works as follows:

1. Start with the original training dataset.
2. Create multiple random samples from the dataset **with replacement**.
3. Train one Decision Tree on each sample.
4. Combine the predictions of all the trees.

Example

Suppose we have:

Original Dataset → 1,000 records

Using bootstrap sampling:

- Dataset 1 → Random 1,000 records
- Dataset 2 → Random 1,000 records
- Dataset 3 → Random 1,000 records

Because sampling happens **with replacement**, the same record can appear multiple times in one sample, while some records may not be selected.

Each bootstrap sample is then used to train a separate Decision Tree.

6. Where Does the “Random” in Random Forest Come From?

Random Forest introduces randomness in **two places**:

1. Random Sampling of Data

Each Decision Tree is trained on a different **bootstrap sample** of the training data.

2. Random Selection of Features

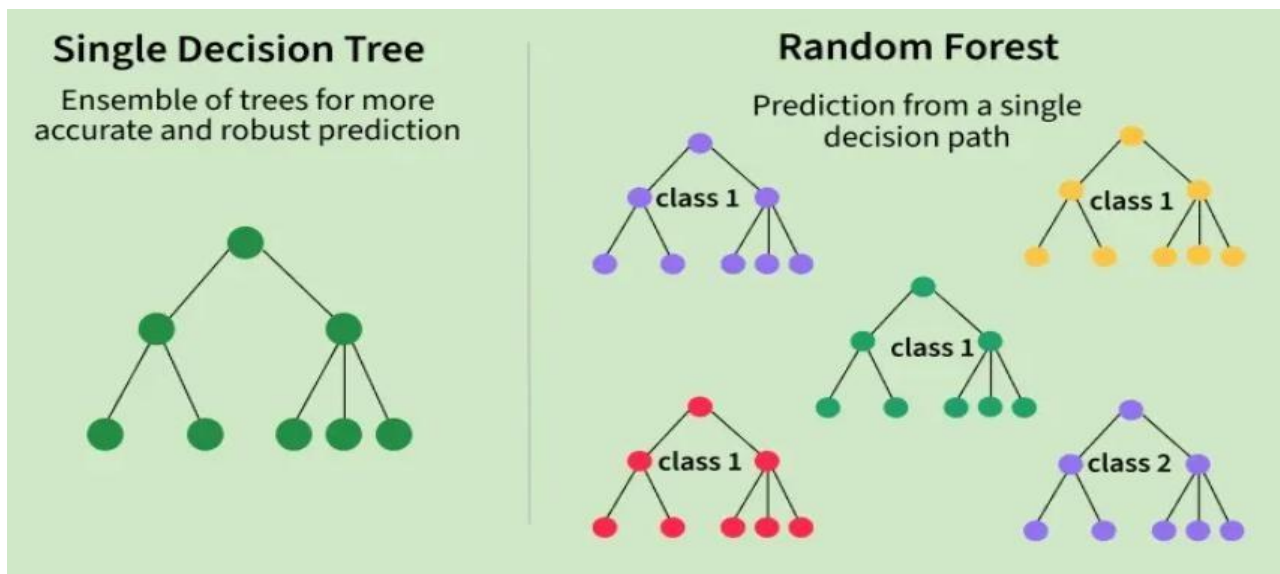
When a Decision Tree looks for the best split, it does **not consider all features**.

Instead:

1. A random subset of features is selected.
2. The best split is found only among those selected features.
3. This process is repeated at every split.

This makes the trees **different from each other**.

Random Forest = Random Data + Random Features + Multiple Decision Trees



7. How Does Random Forest Work?

The overall process can be understood in the following steps:

Step 1: Create Bootstrap Samples

Random Forest creates multiple datasets from the original training data using **sampling with replacement**.

Step 2: Build Multiple Decision Trees

A separate Decision Tree is trained on each bootstrap sample.

Step 3: Select Random Features

At every node, only a random subset of features is considered for finding the best split.

Step 4: Grow the Trees

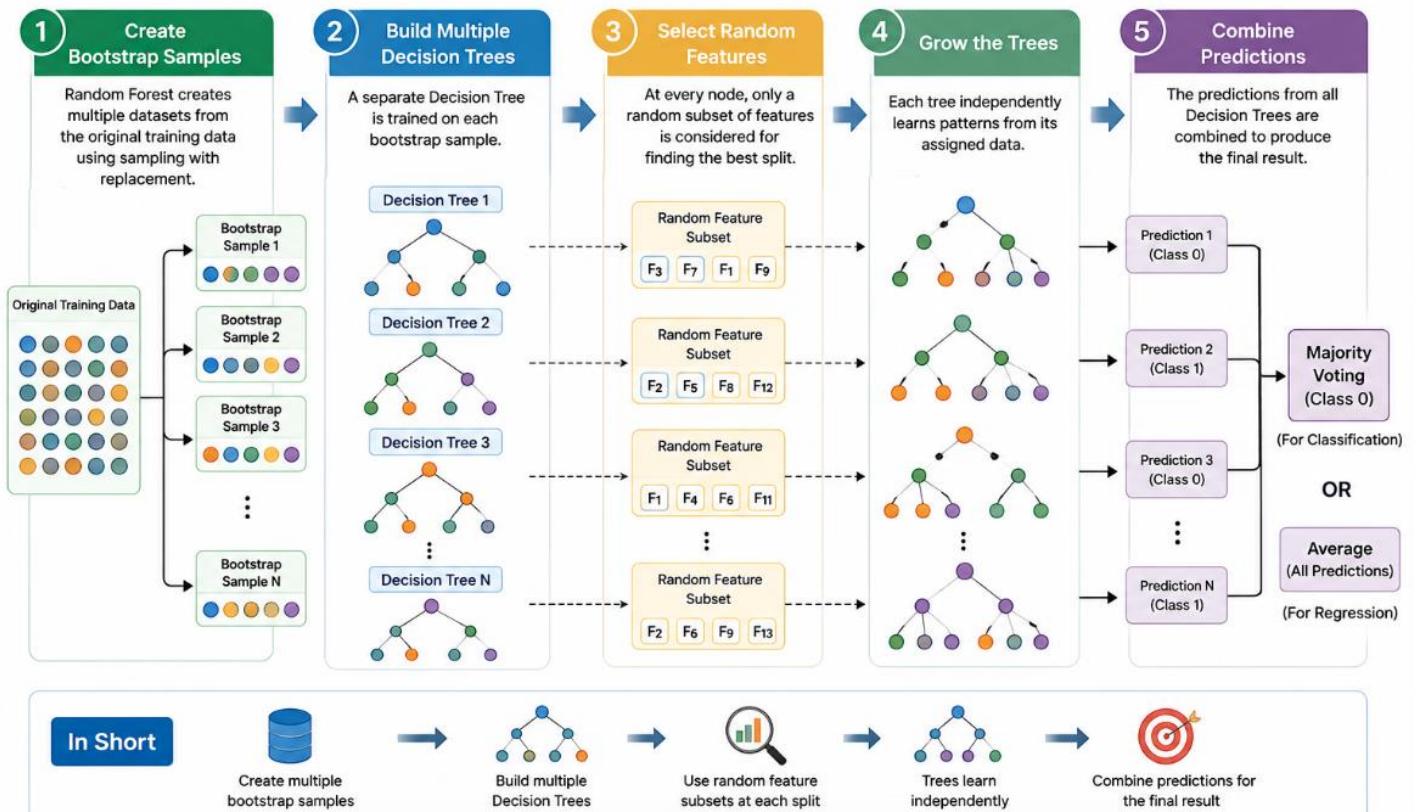
Each tree independently learns patterns from its assigned data.

Step 5: Combine Predictions

The predictions from all Decision Trees are combined to produce the final result.

How Does Random Forest Work?

Random Forest combines multiple Decision Trees to make a more accurate and reliable prediction.



How Does Random Forest Make Classification Predictions?

For a **classification problem**, every Decision Tree predicts a class.

The final prediction is based on **Majority Voting**.

Example

Suppose we have 5 Decision Trees predicting whether a customer will **Churn** or **Not Churn**:

Tree	Prediction
Tree 1	Churn
Tree 2	Not Churn
Tree 3	Churn
Tree 4	Churn
Tree 5	Not Churn

Here:

- Churn = 3 votes
- Not Churn = 2 votes

Therefore:

Final Prediction → Churn

Classification in Random Forest → Majority Voting

How Does Random Forest Make Regression Predictions?

For a **regression problem**, every Decision Tree predicts a numerical value.

The final prediction is calculated by taking the **average of all tree predictions**.

Example

Suppose 5 trees predict house prices:

- Tree 1 → ₹50 Lakhs
- Tree 2 → ₹55 Lakhs
- Tree 3 → ₹52 Lakhs
- Tree 4 → ₹53 Lakhs
- Tree 5 → ₹50 Lakhs

The final prediction is:

Final Prediction = Average of all predictions

Regression in Random Forest → Average Prediction

8. Why Does Random Forest Reduce Overfitting?

A single Decision Tree may make a wrong prediction because it has learned some **noise or specific patterns** from the training data.

However, Random Forest combines predictions from many trees.

Individual trees may make different errors.

When their predictions are combined:

- Random errors tend to cancel out.
- Important patterns are reinforced.
- The final model becomes more stable.

Therefore, Random Forest usually has:

Lower Variance than a single Decision Tree

One Tree may overfit, but many diverse trees combined usually generalize better.

9. Important Hyperparameters of Random Forest

Hyperparameter	What Does It Control?	Effect of Low vs High Value
n_estimators	Number of trees	Low → fewer, faster, less stable; High → more stable, slower
max_depth	Depth of each tree	Low → simpler, may underfit; High → complex, may overfit
min_samples_split	Samples needed to split	Low → more splits, complex tree; High → fewer splits, simpler tree
min_samples_leaf	Minimum samples in a leaf	Low → detailed, may overfit; High → smoother, may underfit
max_features	Features considered at each split	Low → more randomness/diversity; High → less randomness, similar trees
criterion	Quality of a split	Common options → Gini or Entropy
bootstrap	Bootstrap sampling	True → sampling with replacement; False → no bootstrap sampling
random_state	Controls randomness	Same value → reproducible results

10. Advantages of Random Forest

- Random Forest provides very accurate predictions even with large datasets.
- Random Forest can handle missing data well without compromising with accuracy.
- It doesn't require normalization or standardization on dataset.
- When we combine multiple decision trees it reduces the risk of overfitting of the model.

11. Limitations of Random Forest

- It can be computationally expensive especially with a large number of trees.
- It's harder to interpret the model compared to simpler models like decision trees.

12.Q & A

Q1. A Decision Tree has almost 100% training accuracy but much lower testing accuracy. What is happening?

Answer: The tree is likely **overfitting** the training data.

Q2. What happens if we keep increasing max_depth?

Answer: The tree becomes more complex and may eventually **overfit** the training data.

Q3. Why can Random Forest perform better than a single Decision Tree?

Answer: Random Forest combines multiple diverse trees, which can **reduce variance and improve generalization**.

Q4. Why does Random Forest use random subsets of features?

Answer: To create **diversity among the trees**, so that all trees do not make exactly the same decisions.

Q5. If one tree makes an incorrect prediction, will Random Forest necessarily make the same prediction?

Answer: No. Other trees may predict correctly, and majority voting can still produce the correct final prediction.

Q6. Why don't Decision Trees require feature scaling?

Answer: Decision Trees make decisions using **feature thresholds**, not distances or feature magnitudes.

Q7. What happens when `n_estimators` is increased?

Answer: More trees are added to the forest, which generally makes predictions **more stable**, although computation time also increases.

Q8. Why do we use `GridSearchCV`?

Answer: To systematically search for a good combination of **hyperparameters** using **cross-validation**.

Q9. If a Decision Tree gives 100% training accuracy, should we immediately choose it as the best model? Why or why not?

Answer: No. High training accuracy may indicate overfitting. We should also evaluate the model on **unseen test data** and compare its generalization performance.